

BUSFIN 8200: Asset Pricing Foundations

Problem Set 1

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1 Question 1

1.1 Derivation of Equations 1.1–1.3

Throughout, a lowercase letter denotes the log of the corresponding uppercase variable, e.g. $r_{t+1} = \log(R_{t+1})$, $p_t = \log(P_t)$, and $cf_{t+1} = \log(CF_{t+1})$.

Step 1: log return. Start from the definition of the gross return,

$$R_{t+1} = \frac{P_{t+1} + CF_{t+1}}{P_t}. \quad (1)$$

Taking logs,

$$\begin{aligned} \log(R_{t+1}) &= \log(P_{t+1}) - \log(P_t) + \log\left(1 + \frac{CF_{t+1}}{P_{t+1}}\right) \\ \iff r_{t+1} &= p_{t+1} - p_t + \log\left(1 + e^{cf_{t+1} - p_{t+1}}\right). \end{aligned} \quad (2)$$

Step 2: first-order Taylor approximation of the log term. Rearranging (2) so that the polynomial terms are on the left-hand side and the log term is on the right-hand side,

$$r_{t+1} - p_{t+1} + p_t = \log\left(1 + e^{cf_{t+1} - p_{t+1}}\right). \quad (3)$$

Let $x_{t+1} = cf_{t+1} - p_{t+1}$. The function $\log(1 + e^x)$ can be approximated by a first-order Taylor expansion around a point a :

$$\log(1 + e^x) \approx \log(1 + e^a) + \frac{e^a}{1 + e^a} (x - a), \quad (4)$$

which is accurate for small enough $x - a$. The expansion point a is taken to be the time average of $cf_{t+1} - p_{t+1}$, which is assumed to converge to a constant, denoted $a = \overline{cf} - \bar{p}$.

Step 3: substituting the approximation. Applying the Taylor approximation (4) to the right-hand side of (3), with $x = cf_{t+1} - p_{t+1}$ and $a = \overline{cf} - \bar{p}$, and rearranging the terms, we obtain

$$p_t = \kappa_0 + (1 - \kappa) cf_{t+1} - r_{t+1} + \kappa p_{t+1}, \quad (5)$$

where, as in the problem statement,

$$\kappa = \frac{1}{1 + e^a} \quad \text{and} \quad \kappa_0 = -\log(\kappa) - (1 - \kappa) \log\left(\frac{1}{\kappa} - 1\right), \quad (6)$$

with $a = \overline{cf} - \bar{p}$ the expansion point of Step 2 (the problem statement writes this average as $\overline{dp}$).

Step 4: rewriting in terms of the log cash-flow–price ratio. Subtracting cf_t from both sides of (5) and multiplying both sides by -1 , and defining

$$dp_t := cf_t - p_t \quad \text{and} \quad \Delta cf_{t+1} := cf_{t+1} - cf_t, \quad (7)$$

we obtain

$$dp_t = -\kappa_0 + \kappa dp_{t+1} + r_{t+1} - \Delta cf_{t+1}. \quad (8)$$

Step 5: forward substitution. Equation (8) holds at every date, so shifting it one period forward,

$$dp_{t+1} = -\kappa_0 + \kappa dp_{t+2} + r_{t+2} - \Delta cf_{t+2}. \quad (9)$$

Plugging (9) into the dp_{t+1} term on the right-hand side of (8) gives

$$dp_t = -\kappa_0 - \kappa \kappa_0 + \kappa^2 dp_{t+2} + r_{t+1} + \kappa r_{t+2} - \Delta cf_{t+1} - \kappa \Delta cf_{t+2}. \quad (10)$$

Step 6: substitution up to horizon H . Continuing to substitute the future dp terms into (10) up to horizon H , we get

$$dp_t = -\kappa_0 \sum_{h=0}^{H-1} \kappa^h + \kappa^H dp_{t+H} + \sum_{h=1}^H \kappa^{h-1} r_{t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta cf_{t+h}. \quad (11)$$

1.2 Variance decomposition by horizon (data)

1.3 Variance decomposition by horizon (VAR-implied)

1.4 Infinite-horizon decomposition

1.5 Derivation of Equations 1.7–1.9 (Gao and Martin, 2021)

2 Question 2

2.1 Predictive regressions by horizon

Figure 1 plots the adjusted R^2 of the regression

$$\frac{1}{H} \sum_{h=1}^H xR_{e,t+h} = a^{(H)} + b^{(H)} \frac{D_t}{P_t} + \varepsilon_t^{(H)}, \quad H = 1, \dots, 15, \quad (12)$$

estimated by OLS on the EQ Dataset, where $xR_{e,t} = e^{r_{e,t}} - e^{r_{f,t}}$ and $D_t/P_t = e^{dp_t}$. The data are monthly observations of annual variables, so $t+h$ is $12h$ months after t , and each horizon uses every month t for which H years of subsequent data are available (code: `src/q2a.py`).

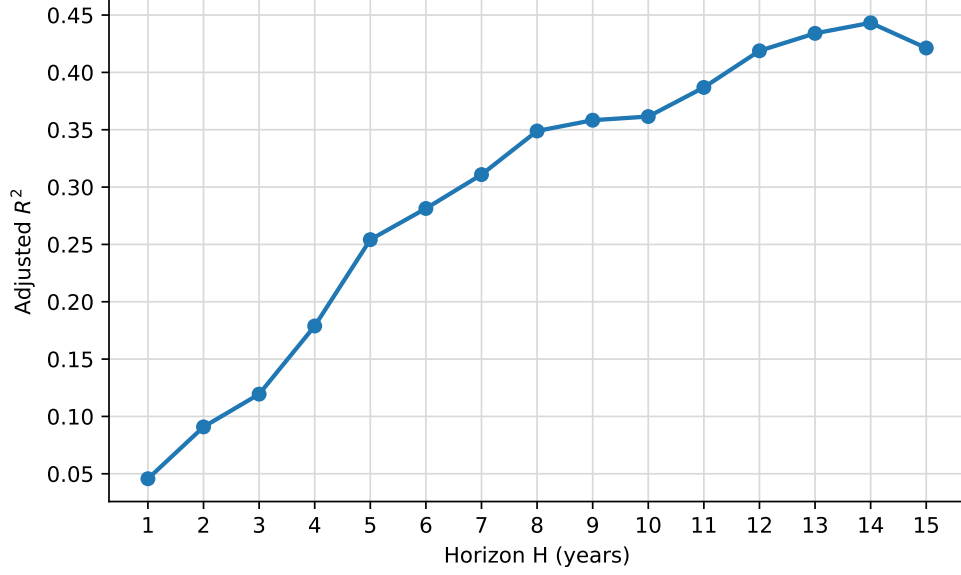


Figure 1: Adjusted R^2 of regression (12) against the horizon H (in years). Monthly observations, 1927:12–2021:12.

2.2 Standard errors for the one-year predictive regression

Table 1 reports the OLS estimates of

$$xR_{e,t+1} = a + b \frac{D_t}{P_t} + \varepsilon_t, \quad (13)$$

together with the standard error and t -statistic of $\hat{b}$ under five standard-error methods. As in Question 2a, $xR_{e,t} = e^{r_{e,t}} - e^{r_{f,t}}$ and $D_t/P_t = e^{d_{p,t}}$; $t+1$ is 12 months after t , and the sample uses every month t for which the subsequent annual return is available (code: `src/q2b.py`).

Standard-error method	SE($\hat{b}$)	$t(\hat{b})$	Lags
(i) OLS	0.3802	7.38	–
(ii) White (1980)	0.6587	4.26	–
(iii) Newey-West (1987), 11 lags	1.2979	2.16	11
(iv) Hansen-Hodrick (1980), 11 lags	1.4521	1.93	11
(v) Newey-West (1987, 1994), automatic lags	1.2311	2.28	24
$\hat{a} = -0.0314$, $\hat{b} = 2.8038$, $T = 1117$ monthly observations			

Table 1: OLS estimates of regression (13) with the standard error and t -statistic of $\hat{b}$ under five standard-error methods. Predictor dates 1927:12–2020:12. “Lags” is the number of autocorrelation lags; for method (v) it is selected by the Newey–West (1994) automatic bandwidth rule.

2.3 Amihud and Hurvich (2004) bias-corrected estimate

Table 2 reports the Amihud and Hurvich (2004) estimate $\hat{b}^c$ of b in regression (13), on the same sample as Table 1. First, D_{t+1}/P_{t+1} is regressed on D_t/P_t by OLS, giving $\hat{\theta}$ and $\hat{\rho}$, and the

slope is bias-corrected as $\hat{\rho}^c = \hat{\rho} + (1 + 3\hat{\rho})/n + 3(1 + 3\hat{\rho})/n^2$ with $n = 94$ years. Second, with $\hat{u}_{t+1}^c = D_{t+1}/P_{t+1} - (\hat{\theta} + \hat{\rho}^c D_t/P_t)$,

$$xR_{e,t+1} = a + b \frac{D_t}{P_t} + b_u \hat{u}_{t+1}^c + \varepsilon_t \quad (14)$$

is estimated by OLS, and $\hat{b}^c$ is the coefficient on D_t/P_t . Its standard error follows Amihud and Hurvich (2004, Eq. 10), $\widehat{SE}(\hat{b}^c) = \sqrt{(\hat{b}_u)^2 (1 + 3/n + 9/n^2)^2 \widehat{\text{Var}}(\hat{\rho}) + \widehat{SE}(\hat{b}^c)^2}$ with $n = 94$ (code: `src/q2c.py`).

Standard error of $\hat{b}^c$	$\text{SE}(\hat{b}^c)$	$t(\hat{b}^c)$
(i-a) Eq. (10), OLS ingredients	0.3876	6.03
(i-b) Eq. (10), Newey-West (1987) 11 lags ingredients	1.5170	1.54
$\hat{b}^c = 2.3362, \quad \hat{a} = -0.0314, \quad \hat{b}_u = -13.4837$		
$\hat{\rho} = 0.7197, \quad \hat{\rho}^c = 0.7544, \quad n = 94 \text{ years}, \quad T = 1117 \text{ monthly observations}$		
OLS $\hat{b}$ (Question 2b) = 2.8038		

Table 2: Amihud and Hurvich (2004) estimate of b in regression (13) via the augmented regression (14). In row (i-a), $\widehat{\text{Var}}(\hat{\rho})$ and $\widehat{SE}(\hat{b}^c)$ are plain OLS estimates; in row (i-b), both are Newey–West (1987) estimates with 11 lags. Predictor dates 1927:12–2020:12.

2.4 Out-of-sample estimation (expanding window)

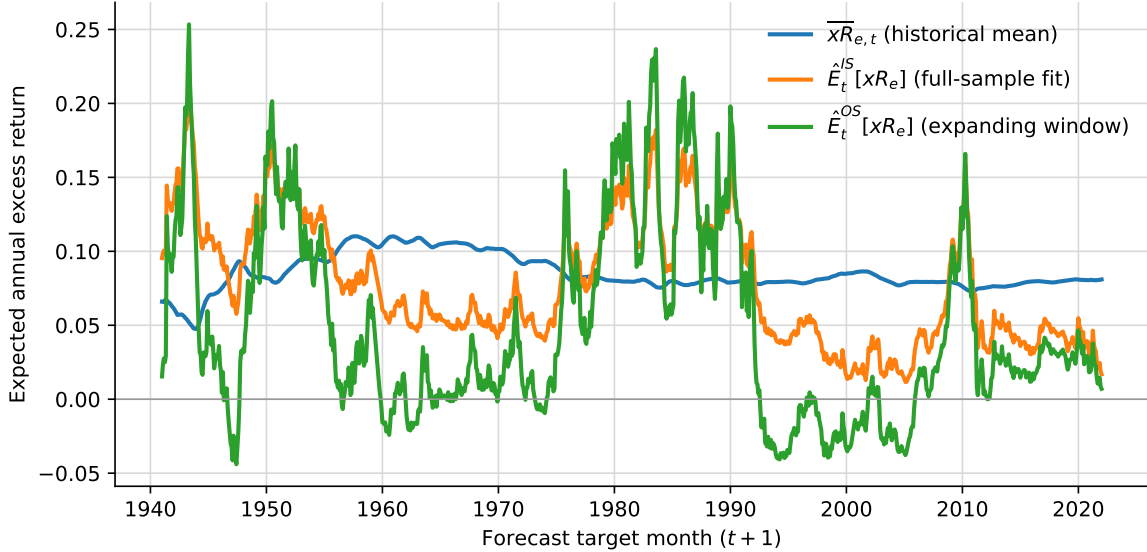


Figure 2: Expanding-window historical mean $\overline{xR}_{e,t}$, in-sample fitted values $\hat{E}_t^{IS}[xR_e]$ from the full-sample regression of Question 2b, and out-of-sample forecasts $\hat{E}_t^{OS}[xR_e]$ from regression (13) estimated with data available at t , plotted against the forecast-target month $t + 1$, 1940:12–2021:12.

The out-of-sample R^2 over the forecast-target months 1940:12–2021:12 is $R_{OS}^2 = -0.0135$.

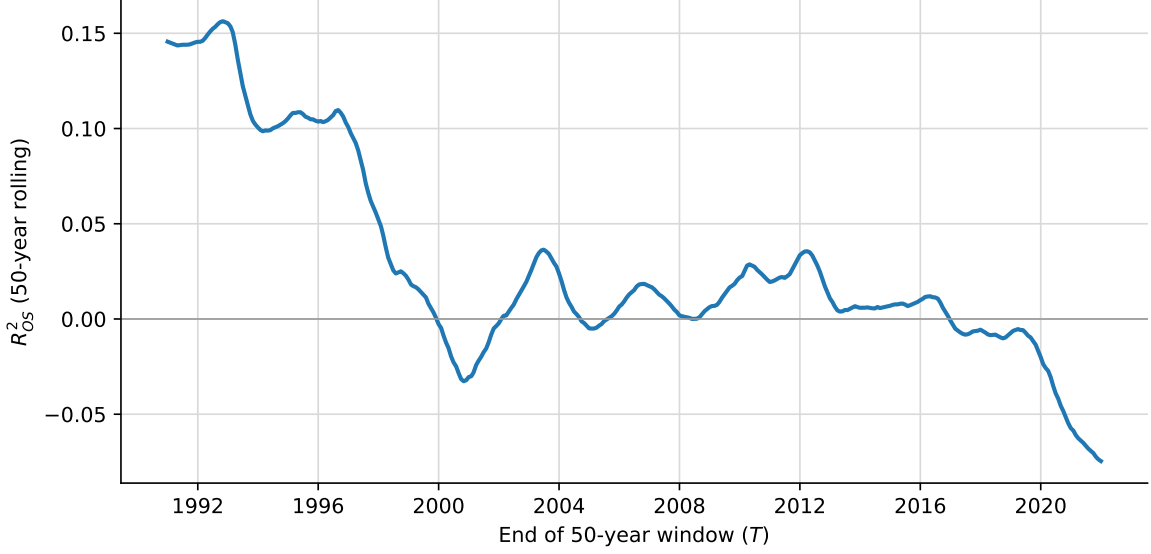


Figure 3: R^2_{OS} computed over a rolling window of 600 forecast-target months ending at T , for T from 1990:12 to 2021:12, with $\hat{a}_t$, $\hat{b}_t$ and $\overline{xR}_{e,t}$ unchanged.

2.5 Out-of-sample estimation with steady-state restrictions

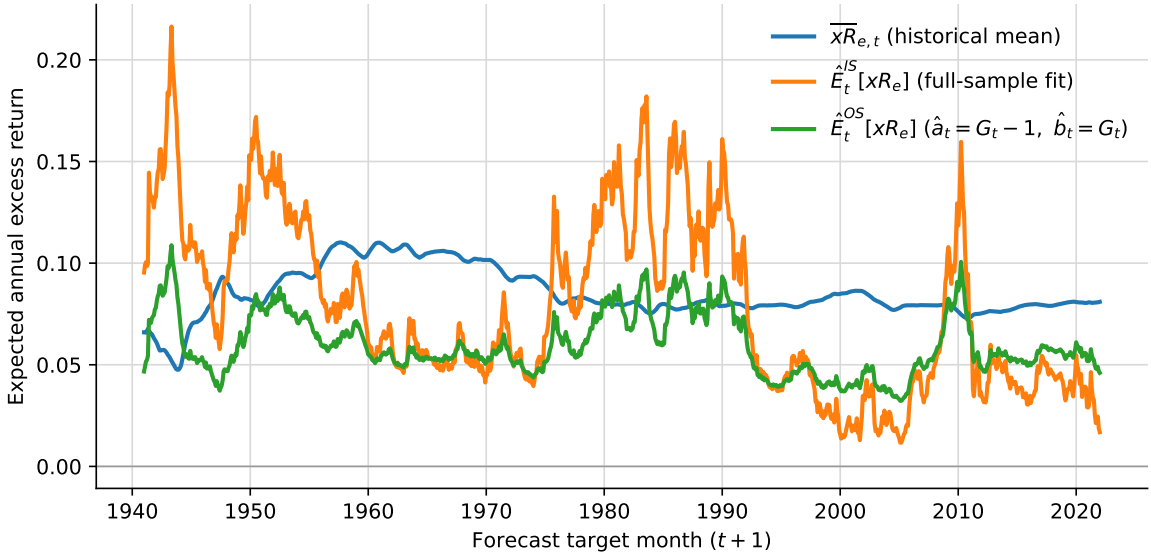


Figure 4: Expanding-window historical mean $\overline{xR}_{e,t}$, in-sample fitted values $\hat{E}_t^{IS}[xR_e]$ from the full-sample regression of Question 2b, and out-of-sample forecasts $\hat{E}_t^{OS}[xR_e] = (G_t - 1) + G_t D_t/P_t$, where G_t is the expanding mean of $e^{\Delta d}$ over 1927:12 to t , plotted against the forecast-target month $t + 1$, 1940:12–2021:12.

The out-of-sample R^2 over the forecast-target months 1940:12–2021:12 is $R^2_{OS} = 0.0294$.

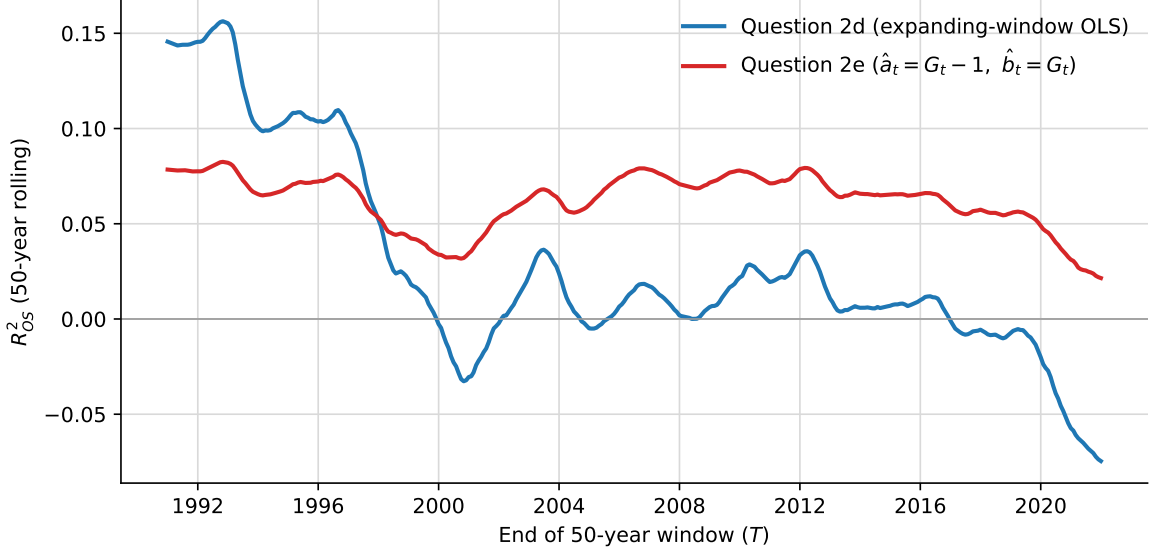


Figure 5: R_{OS}^2 over a rolling window of 600 forecast-target months ending at T , for T from 1990:12 to 2021:12, for the expanding-window OLS forecasts of Question 2d and the restricted forecasts of Question 2e ($\hat{a}_t = G_t - 1$, $\hat{b}_t = G_t$).

3 Question 3

3.1 Momentum signal construction and validation

The momentum signal of stock j at the end of month τ is

$$MOM_{j,\tau} = \prod_{m=\tau-11}^{\tau-1} (1 + R_{j,m}) - 1, \quad (15)$$

the cumulative return from the end of month $\tau - 12$ to the end of month $\tau - 1$, where $R_{j,m}$ is the CRSP monthly return (`MthRet`, which includes delisting returns). The returns are taken from every month of the CRSP file for the same PERMNO, whether or not the stock is in the universe in those months, and $MOM_{j,\tau}$ is missing when any of the 11 months has a missing return or no observation.

The signal months are 1963:06–2024:12. The universe at τ consists of common stocks of firms incorporated in the United States (CRSP fields `ShareType` = NS, `SecurityType` = EQTY, `SecuritySubType` = COM, `USIncFlg` = Y, `IssuerType` $\in$ {ACOR, CORP}, the equivalent of share codes 10 and 11) trading on NYSE, Amex or Nasdaq (`PrimaryExch` $\in$ {N, A, Q}, `ConditionalType` = RW, `TradingStatusFlg` = A), excluding utilities ($4900 \leq SIC \leq 4949$) and financials ($6000 \leq SIC \leq 6999$) and requiring non-missing market equity at τ . The SIC code is the CRSP SICCD at τ ; a code of zero is treated as missing and replaced by the most recent earlier non-zero code of the same PERMNO or, if there is none, by its next later one, and a stock that never has a non-zero code is kept.

For each month τ , the cross-firm regression

$$MOM_{j,\tau}^{CZ} = a_\tau + b_\tau MOM_{j,\tau} + \varepsilon_{j,\tau} \quad (16)$$

is estimated by OLS, where MOM^{CZ} is Mom12m from the CZ Dataset, merged on the stock identifier and the month τ , using the stocks in the universe at τ for which both signals are non-missing (2,428,767 stock-month observations in 739 months). Figures 6–8 plot $\hat{a}_\tau$, $\hat{b}_\tau$ and the R^2 of these regressions, and Table 3 summarizes them (code: `src/q3a.py`, `src/q3_common.py`).

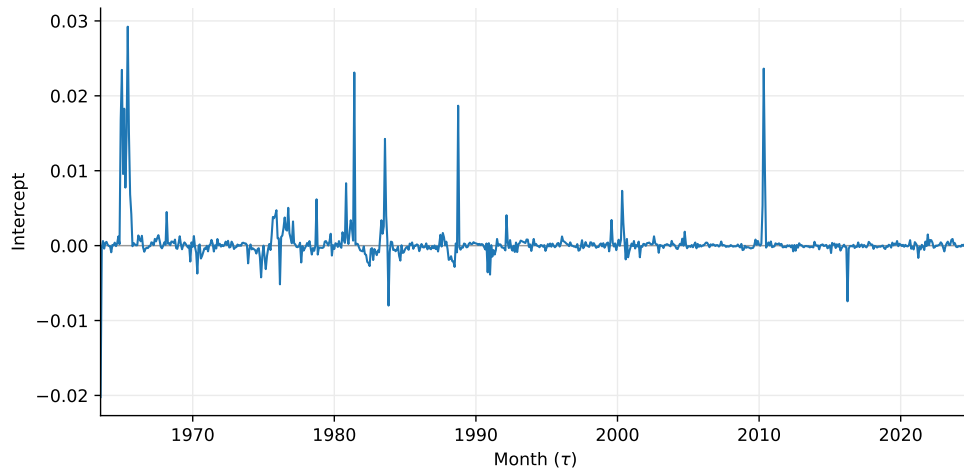


Figure 6: Intercept $\hat{a}_\tau$ of the monthly cross-firm regression (16), 1963:06–2024:12. The grey line marks 0.

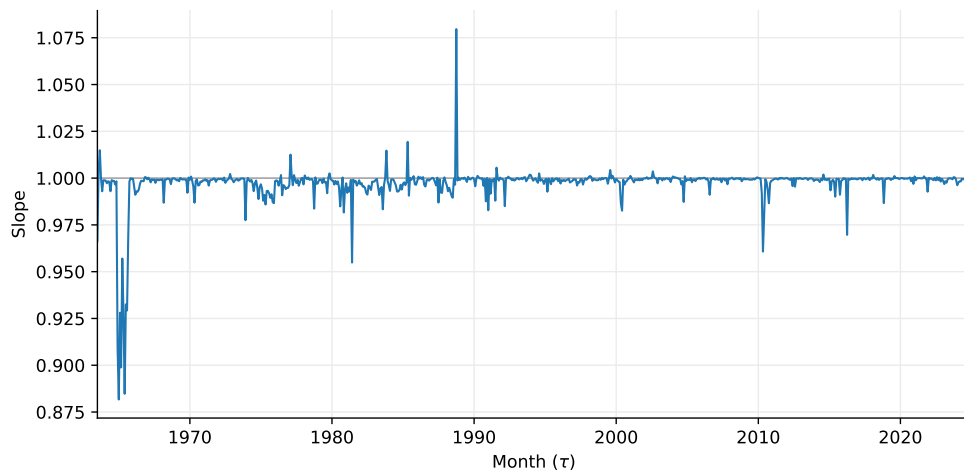


Figure 7: Slope $\hat{b}_\tau$ of the monthly cross-firm regression (16), 1963:06–2024:12. The grey line marks 1.

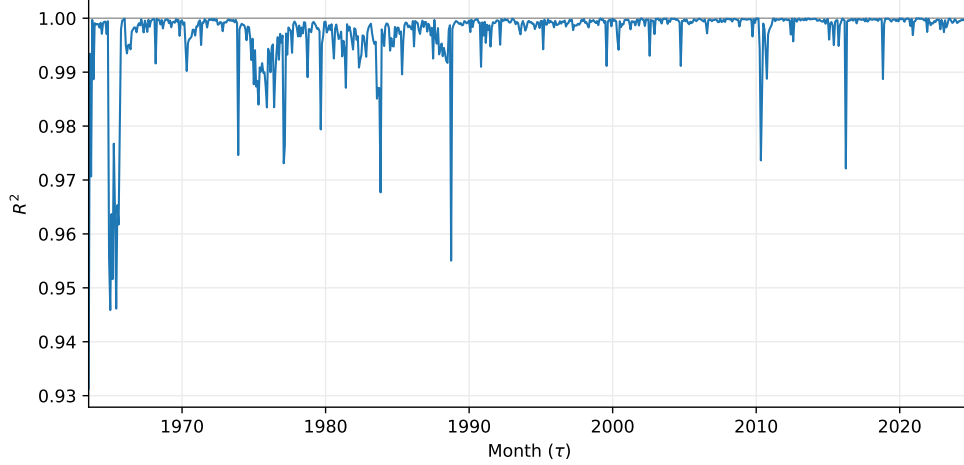


Figure 8: R^2 of the monthly cross-firm regression (16), 1963:06–2024:12. The grey line marks 1.

	Mean	Minimum	Maximum
Intercept	0.0003	-0.0202	0.0292
Slope	0.9972	0.8817	1.0796
R^2	0.9973	0.9313	1.0000
N	3,287	1,655	5,350

Table 3: Time-series mean, minimum and maximum of the intercept, slope, R^2 and number of stocks N of the monthly cross-firm regressions (16) of MOM^{CZ} on MOM , 739 signal months from 1963:06 to 2024:12.

3.2 Book-to-market signal construction and validation

The book-to-market signal of stock j for year t is

$$BM_{j,t} = \frac{BE_{j,t-1}}{ME_{j,t-1}}, \quad (17)$$

where $BE_{j,t-1}$ is the book equity from the COMPUSTAT record whose fiscal year ends in calendar year $t - 1$ and $ME_{j,t-1} = |PRC| \times SHROUT$ is the CRSP market equity of the stock in December of year $t - 1$ (book equity is in millions and market equity in thousands of dollars, so the ratio is multiplied by 1000). $BM_{j,t}$ is assigned to the months τ from June of year t to May of year $t + 1$, for $t = 1963, \dots, 2024$, so the signal months are 1963:06–2024:12; the universe at τ is the one of Question 3a. $BM_{j,t}$ is missing when the stock has no CRSP observation or no market equity in December of $t - 1$.

Book equity is $BE = SE + TXDITC - BVPS$, where SE is SEQ, else CEQ + PSTK, else AT - LT; $TXDITC$ is TXDITC, or 0 when it is missing; and $BVPS$ is PSTKRV, else PSTKL, else PSTK, else 0. Observations with unavailable SE or with $BE \leq 0$ are excluded.

The accounting data come from the CRSP/COMPUSTAT Merged file, with link types LC and LU, primary links (LINKPRIM P or C), the fiscal year-end date inside the link window, and records reported in US dollars. When a stock has several records with a fiscal year-end in the same calendar year, a P link is preferred to a C link, and then the latest fiscal year-end date is used. To avoid

backfilling bias, the record used must be preceded by at least two earlier records (distinct fiscal year-end dates) of the same firm in the file.

In the release of the CZ Dataset used here (2025.10), **BMdec** is the book-to-market ratio in levels and not its logarithm, so $BM^{CZ} = \text{BMdec}$ is used instead of $\exp(\text{BMdec})$. For each month τ , the cross-firm regression

$$BM_{j,\tau}^{CZ} = a_\tau + b_\tau BM_{j,\tau} + \varepsilon_{j,\tau} \quad (18)$$

is estimated by OLS, with BM^{CZ} merged on the stock identifier and the month τ , using the stocks in the universe at τ for which both signals are non-missing (1,803,835 stock-month observations in 739 months; between 454 and 3,914 stocks per month). Figures 9–11 plot $\hat{a}_\tau$, $\hat{b}_\tau$ and the R^2 of these regressions (code: `src/q3b.py`, `src/q3_common.py`).

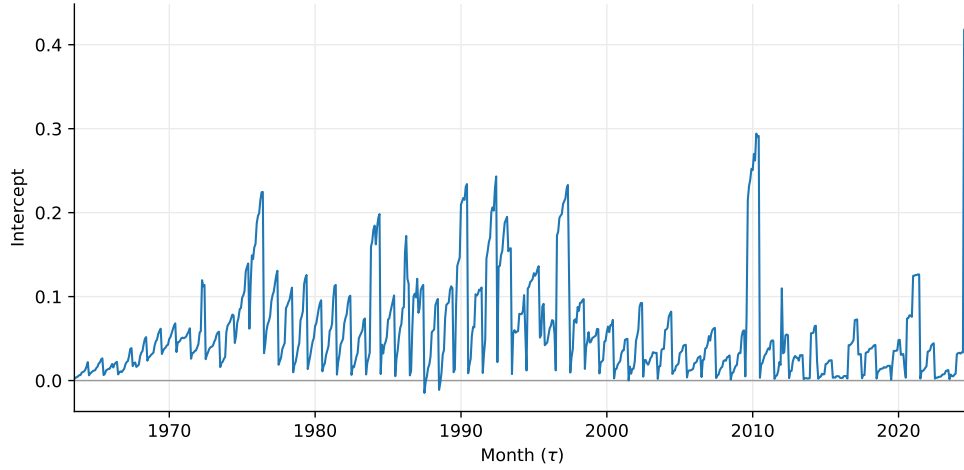


Figure 9: Intercept $\hat{a}_\tau$ of the monthly cross-firm regression (18), 1963:06–2024:12. The grey line marks 0.

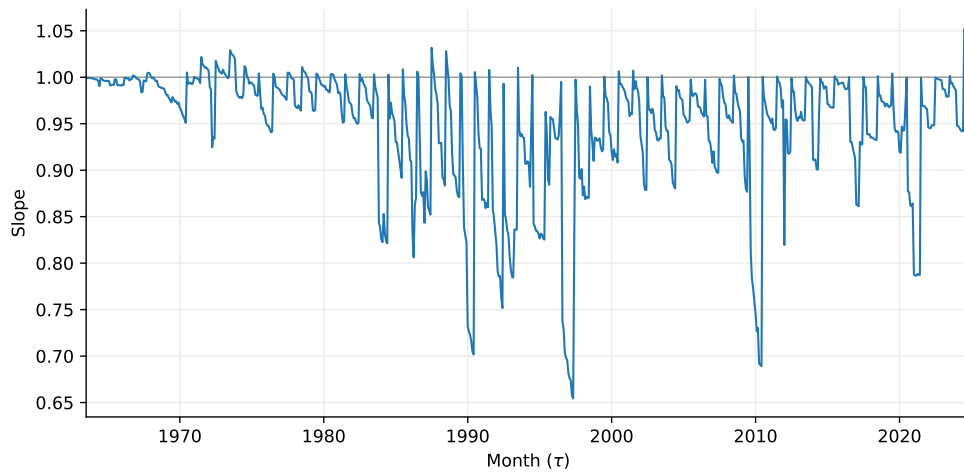


Figure 10: Slope $\hat{b}_\tau$ of the monthly cross-firm regression (18), 1963:06–2024:12. The grey line marks 1.

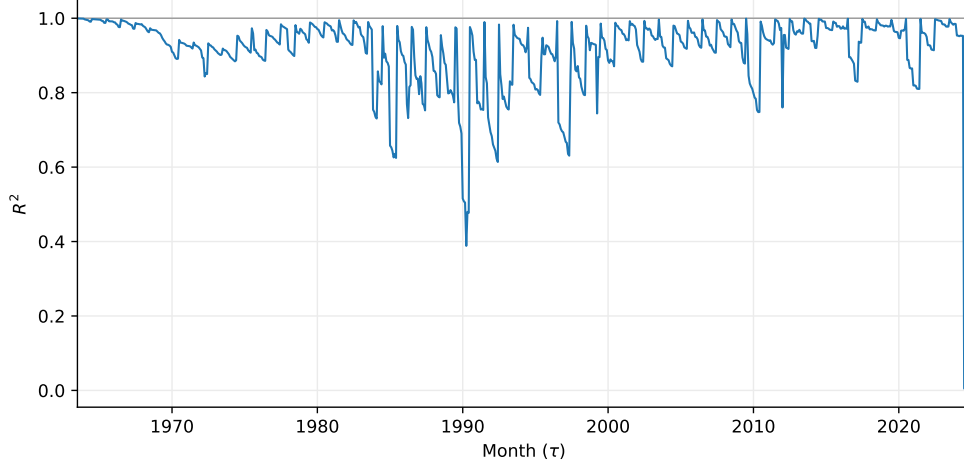


Figure 11: R^2 of the monthly cross-firm regression (18), 1963:06–2024:12. The grey line marks 1.

3.3 Decile portfolios

Decile portfolios are formed on each of the three signals of the CZ Dataset, $BM^{CZ} = \text{BMdec}$ (in levels, as in Question 3b), $MOM^{CZ} = \text{Mom12m}$ and $GP^{CZ} = \text{GP}$, dated τ . A stock enters a sort if it is in the universe at the formation date and its signal is non-missing and finite; for BM^{CZ} , stocks with $\text{BMdec} \leq 0$ are excluded.

With annual rebalancing, the portfolios are formed at the end of June of year t with the signal and the universe of June t and are held for the return months from July of year t to June of year $t + 1$; a stock without a signal in June t is left out until the next June. With monthly rebalancing, the portfolios are formed at the end of month τ with the signal and the universe of τ and are held for the return month $\tau + 1$. The breakpoints are the 10th, 20th, ..., 90th percentiles of the signal at the formation date, computed among the NYSE stocks in the sort (NYSE breakpoints) or among all stocks in the sort (general breakpoints); a stock is in decile d if its signal is above the $(d - 1)$ -th breakpoint and at or below the d -th, so that decile 1 has the lowest and decile 10 the highest signal.

The portfolio return of month $\tau + 1$ uses the members with a non-missing return in $\tau + 1$, with equal weights (EW) or with weights proportional to the market equity at the end of month τ (VW), which are updated every month, also within the annual holding period; a member with missing market equity at τ is excluded from the VW return of $\tau + 1$. Excess returns are $xR_{p,\tau+1} = R_{p,\tau+1} - R_{f,\tau+1}$, with the risk-free return from the Ken French data library. The five types of portfolios are: (i) VW, annual rebalancing, NYSE breakpoints; (ii) EW, annual rebalancing, NYSE breakpoints; (iii) VW, monthly rebalancing, NYSE breakpoints; (iv) VW, annual rebalancing, general breakpoints; (v) EW, monthly rebalancing, general breakpoints. The return months are 1963:07–2025:01 (739 months) for all five types; the last annual formation is June 2024 (code: `src/q3c.py`, `src/q3_common.py`).

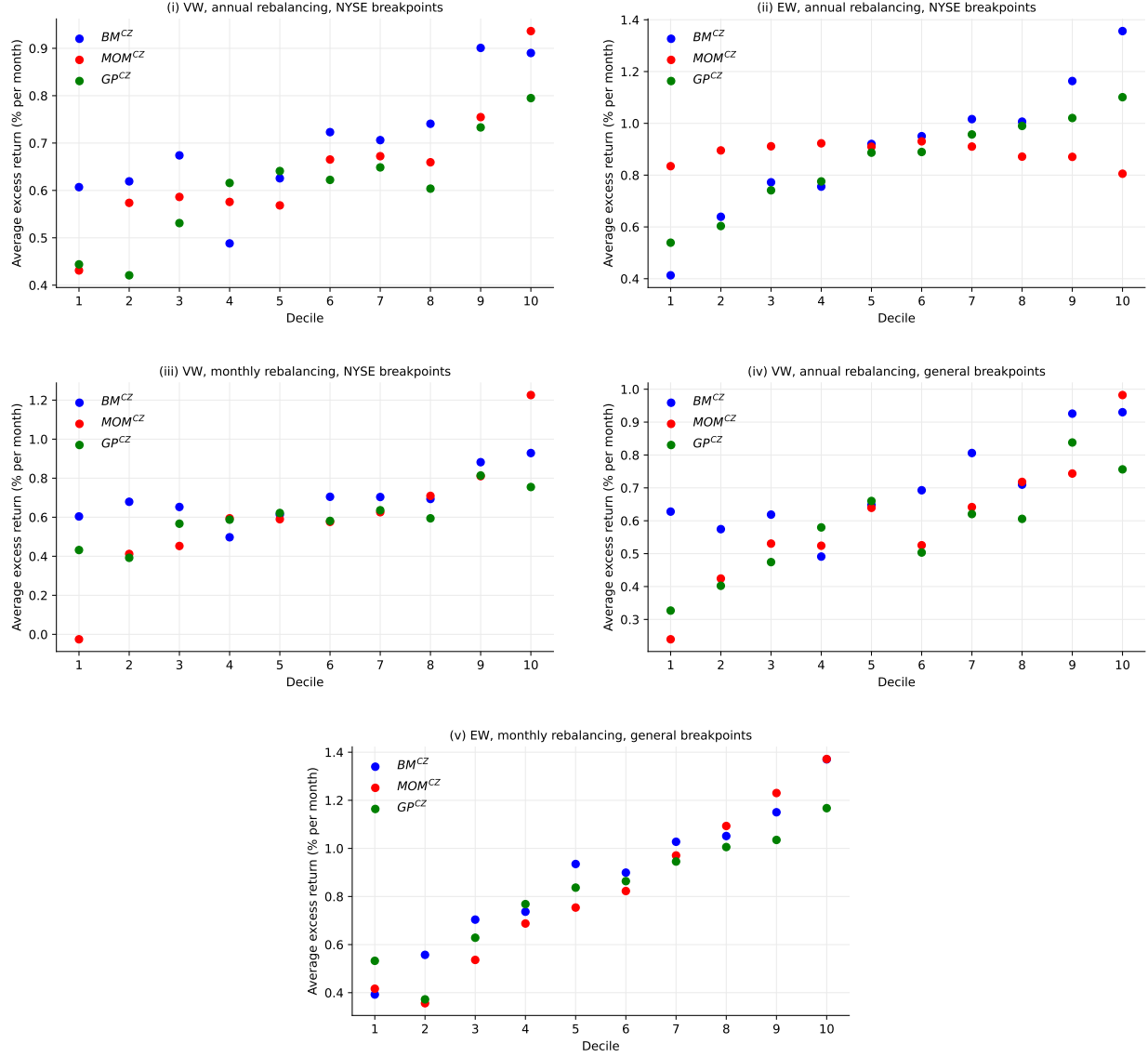


Figure 12: Average monthly excess return (in percent) of the decile portfolios against the decile, for the five types of portfolios; blue: BM^{CZ} , red: MOM^{CZ} , green: GP^{CZ} . Return months 1963:07–2025:01.

Table 4 reports, for each type and signal, the average return of the HML portfolio, defined as decile 10 minus decile 1.

Type	Portfolios	BM^{CZ}	MOM^{CZ}	GP^{CZ}
(i)	VW, annual, NYSE breakpoints	0.283 [1.27]	0.505 [2.19]	0.351 [2.10]
(ii)	EW, annual, NYSE breakpoints	0.943 [4.18]	-0.029 [-0.14]	0.562 [3.29]
(iii)	VW, monthly, NYSE breakpoints	0.325 [1.51]	1.251 [4.62]	0.323 [1.89]
(iv)	VW, annual, general breakpoints	0.302 [1.13]	0.743 [2.58]	0.429 [2.05]
(v)	EW, monthly, general breakpoints	0.978 [4.12]	0.955 [3.15]	0.635 [2.98]

Table 4: Average returns of the HML portfolios (decile 10 minus decile 1), in percent per month, with t -statistics in brackets. The t -statistics come from a regression of the HML return on a constant with Newey–West (1987, 1994) standard errors, with the lag length selected by the automatic bandwidth rule as in method (v) of Question 2b (between 2 and 15 lags). Return months 1963:07–2025:01 (739 observations for each portfolio).

3.4 Fama–MacBeth regressions

3.5 Portfolio-level multivariate regressions

4 Question 4

4.1 Yields, forward rates, and returns

From the Fama–Bliss yields $Y_{b,t}^{(H)}$ (in percent) for $H = 1, \dots, 5$, the log yields, log forward rates and log annual returns are

$$y_{b,t}^{(H)} = \log\left(1 + Y_{b,t}^{(H)}/100\right), \quad f_{b,t}^{(H)} = H y_{b,t}^{(H)} - (H-1) y_{b,t}^{(H-1)}, \quad r_{b,t}^{(H)} = H y_{b,t-1}^{(H)} - (H-1) y_{b,t}^{(H-1)}, \quad (19)$$

with $y^{(0)} = 0$, so that $f_{b,t}^{(1)} = y_{b,t}^{(1)}$ and $r_{b,t}^{(1)} = y_{b,t-1}^{(1)}$; $t-1$ is 12 months before t . The spreads are $xy^{(H)} = y^{(H)} - y^{(1)}$, $xf^{(H)} = f^{(H)} - f^{(1)}$ and $xr^{(H)} = r^{(H)} - r^{(1)}$ (code: `src/q4a.py`).

Average (%)	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$xy_{b,t}^{(H)}$ (yield spread)	0.169	0.331	0.471	0.568
$xf_{b,t}^{(H)}$ (forward spread)	0.339	0.654	0.892	0.956
$xr_{b,t}^{(H)}$ (excess return)	0.315	0.607	0.820	0.872
Common sample 1953:06–2024:12				

Table 5: Average yield spreads, forward spreads and excess log returns of H -year Fama–Bliss discount bonds over the 1-year bond, $H = 2, \dots, 5$, in percent per year. Monthly observations over the common sample on which all three series are defined, 1953:06–2024:12.

4.2 Hold-to-maturity excess return regressions

Table 6 reports OLS estimates of

$$\frac{1}{H} xr_{b,t \rightarrow t+H}^{(H)} = a^{(H)} + b^{(H)} xy_{b,t}^{(H)} + \varepsilon_t^{(H)}, \quad H = 2, \dots, 5, \quad (20)$$

where the hold-to-maturity excess return is

$$xr_{b,t \rightarrow t+H}^{(H)} = \sum_{h=1}^H xr_{b,t+h}^{(H-h+1)} = \sum_{h=1}^H \left(r_{b,t+h}^{(H-h+1)} - r_{b,t+h}^{(1)} \right), \quad (21)$$

with r and xr as in (19), $t+h$ being $12h$ months after t , and the $h=H$ term identically zero because $xr_{b,t}^{(1)} = 0$. Each regression uses every month t for which all non-zero terms of (21) are observed, i.e. $t + 12(H-1) \leq 2024:12$. The t -statistics use Hansen–Hodrick (1980) standard errors with $L = 12H - 1$ lags (code: `src/q4b.py`).

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$b^{(H)}$	0.677	0.532	0.414	0.345
	[3.56]	[3.14]	[2.46]	[2.25]
R^2 (%)	7.7	5.2	3.8	3.2

Table 6: OLS estimates of $b^{(H)}$ in regression (20), with Hansen–Hodrick (1980) t -statistics ($12H - 1$ lags) in brackets and R^2 in percent. Monthly observations from 1952:06; the last predictor month is 2023:12, 2022:12, 2021:12 and 2020:12 for $H = 2, 3, 4, 5$ (859, 847, 835 and 823 observations).

4.3 One-year excess return regressions on forward spreads

Table 7 reports OLS estimates of

$$xr_{b,t+1}^{(H)} = a^{(H)} + b^{(H)} xf_{b,t}^{(H)} + \varepsilon_t^{(H)}, \quad H = 2, \dots, 5, \quad (22)$$

with xr and xf as in (19) and $t+1$ being 12 months after t . Each regression uses every month t for which $xr_{b,t+1}^{(H)}$ is observed, i.e. $t + 12 \leq 2024:12$, so all four samples are 1952:06–2023:12. The t -statistics use Newey–West (1987) standard errors with the lag length selected by the Newey–West (1994) automatic bandwidth rule, as in method (v) of Question 2b (code: `src/q4c.py`).

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$b^{(H)}$	0.677	0.889	1.116	0.964
	[3.23]	[3.35]	[3.63]	[2.92]
R^2 (%)	7.7	8.6	10.7	6.7

Table 7: OLS estimates of $b^{(H)}$ in regression (22), with Newey–West (1987, 1994) t -statistics in brackets (selected lags 23, 23, 24 and 23 for $H = 2, 3, 4, 5$) and R^2 in percent. Monthly observations, predictor months 1952:06–2023:12 (859 observations for each H).

4.4 Cochrane–Piazzesi factor

4.5 Excess return regressions on the CP factor